

Exact Product-Measure Constants in a Question on Ramsey-Type Spheres

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Status

This is a partial solution packet for arXiv:2308.07668, “On Ramsey-type properties of the distance in nonseparable spheres” by Piotr Koszmider. It settles the exact values of the constants c_n and c appearing in Proposition 55 of the source paper. The endpoint dichotomy question for $S_{L_p(\{0,1\}^\kappa)}$ remains separate.

The last finite case, $1 < p < 2$, is obtained by identifying the relevant finite atom inequality with the largest Rayleigh quotient of the complete graph p -Laplacian. The complete graph spectrum is due to Amghibech, and is used here in the form quoted in Zhang’s Section 4.3 [3].

Source Evidence

The following crop is from page 25 of the source PDF.

and $(\varphi_B \circ \varphi_A^{-1})(x_1, \dots, x_n) = \varphi_{A \cap B}(x_1, \dots, x_n)$ as the functions characterizing A and B in the increasing order agree on $A \cap B$ as they agree with e_l . So $\|f - f'\|_{L_p}$ is not bigger than c_n by the definition of c_n , as required. $\square$

We do not know the exact value of c_n or of c in the above proof or if S_{L_p} is dichotomous. Note that $k^+(S_{L_p})$ is not $\sqrt[p]{2}$ in general because we can construct in $L_p(\{0,1\}^{\omega_1})$ uncountable sets of independent variables assuming values 1 and -1 with equal probability $1/2$. They have norm 1 but their differences have the norm equal to $2^{1-1/p}$ which is strictly bigger than $\sqrt[p]{2}$ if p is sufficiently large.

The relevant source sentence is:

“We do not know the exact value of c_n or of c in the above proof or if S_{L_p} is dichotomous.”

Source Notation

Let μ be the product probability measure on $\{0,1\}^\kappa$, where $\kappa \leq 2^\omega$ is uncountable. In Proposition 55 of the source paper, for $A \in [\kappa]^n$, the map T_A sends a function $F \in L_p(\{0,1\}^n)$ to the function on $\{0,1\}^\kappa$ depending on the coordinates in A . The constant c_n is the supremum of

$$\|T_A(F) - T_B(F)\|_{L_p}$$

over all unit vectors $F \in L_p(\{0,1\}^n)$ and all admissible pairs $A, B \in [\kappa]^n$ whose common coordinates occur in the same positions. Finally $c = \sup_n c_n$.

Proof Intuition

After the common coordinates of A and B are fixed, the two random variables $T_A(F)$ and $T_B(F)$ are independent copies of the same smaller finite random variable. Thus the problem is controlled by finite complete graph quotients

$$\sup_{x \neq 0} \frac{\sum_{i,j} |x_i - x_j|^p}{\sum_i |x_i|^p}.$$

For $1 < p < 2$, the largest quotient is not the limiting negative-type value: the finite atom mass forces a strict gap. Amghibech's spectrum for the complete graph p -Laplacian gives exactly this gap, and the largest eigenvalue is the one corresponding to one positive atom, one negative atom, and all other atoms zero. For $p > 2$, the same spectrum is maximized by a balanced positive/negative split; because 2^n is even, this gives the Rademacher value at every finite level.

Lemma 1 (complete graph quotient). *Let $N \geq 2$, and for $1 < p < \infty$ define*

$$M_{N,p} := \sup_{x \in \mathbb{R}^N \setminus \{0\}} \frac{\sum_{i,j=1}^N |x_i - x_j|^p}{\sum_{i=1}^N |x_i|^p}.$$

Then

$$M_{N,p} = 2N - 4 + 2^p, \quad 1 < p \leq 2.$$

If $2 \leq p < \infty$ and N is even, then

$$M_{N,p} = N2^{p-1}.$$

Proof. For $p = 2$, the identity

$$\sum_{i,j} |x_i - x_j|^2 = 2N \sum_i |x_i - \bar{x}|^2 \leq 2N \sum_i |x_i|^2$$

gives $M_{N,2} = 2N$, and both displayed formulas agree with this value.

Assume first that $p \neq 2$. Let K_N be the complete graph on N vertices. Zhang quotes Amghibech's Theorem 6 as follows: for $N \geq 3$, the nonzero normalized p -Laplacian eigenvalues of K_N are

$$\lambda_{i,j} = \frac{1}{N-1} \left(N - i - j + \left(i^{1/(p-1)} + j^{1/(p-1)} \right)^{p-1} \right), \quad i, j \in \mathbb{Z}_+, \quad i + j \leq N.$$

Here the normalized Rayleigh quotient is

$$F_p(x) = \frac{\sum_{\{i,j\} \in E(K_N)} |x_i - x_j|^p}{(N-1) \sum_i |x_i|^p}.$$

Since F_p attains its maximum on the compact ℓ_p -sphere and a maximizer is a critical point, $\sup F_p$ is the largest value among the $\lambda_{i,j}$. Also

$$M_{N,p} = 2(N-1) \sup F_p,$$

because $M_{N,p}$ uses ordered pairs while F_p uses unordered edges.

Put $q = 1/(p-1)$. If $1 < p < 2$, then $q > 1$. The expression to maximize is

$$H(i, j) = N - i - j + \|(i, j)\|_q.$$

For $i, j \geq 1$, the quantity $i + j - \|(i, j)\|_q$ is strictly increasing in each variable, since

$$\frac{\partial}{\partial i} \left(i + j - (i^q + j^q)^{1/q} \right) = 1 - \left(\frac{i^q}{i^q + j^q} \right)^{(q-1)/q} > 0.$$

Thus the maximum is attained at $i = j = 1$, and

$$\sup F_p = \frac{N - 2 + 2^{p-1}}{N - 1}.$$

Hence $M_{N,p} = 2N - 4 + 2^p$.

If $p > 2$, then $0 < q < 1$. For a fixed $s = i + j$, concavity of t^q implies that $(i^q + j^q)^{1/q}$ is maximized when i and j are as balanced as possible. When N is even, taking $s = N$ and $i = j = N/2$ gives

$$H(i, j) = (2(N/2)^q)^{1/q} = N2^{p-2}.$$

For any $s \leq N$, the same balanced estimate gives

$$H(i, j) \leq N - s + 2^{p-2}s,$$

which is maximized at $s = N$ because $2^{p-2} > 1$. Therefore $\sup F_p = N2^{p-2}/(N - 1)$, and $M_{N,p} = N2^{p-1}$.

For $N = 2$, the same values follow directly from the two-point quotient, so the proof is complete. $\square$

Lemma 2 (conditioning reduction). *Let $N = 2^n$. For the source constants c_n ,*

$$c_n^p \leq \frac{M_{N,p}}{N}.$$

Proof. Fix $F \in L_p(\{0, 1\}^n)$ and an admissible pair A, B . Suppose that $r = |A \cap B|$, with common coordinates in the same positions, and condition on these r common coordinates. Given the common coordinates, $T_A(F)$ and $T_B(F)$ are iid copies of the same function on a cube with 2^{n-r} atoms. Lemma 1 applied to this smaller finite cube gives

$$\mathbb{E}(|T_A(F) - T_B(F)|^p \mid A \cap B) \leq d_{n-r,p} \mathbb{E}(|T_A(F)|^p \mid A \cap B),$$

where

$$d_{m,p} = \begin{cases} 2 - (4 - 2^p)/2^m, & 1 < p \leq 2, \ m \geq 1, \\ 2^{p-1}, & 2 \leq p < \infty, \ m \geq 1, \\ 0, & m = 0. \end{cases}$$

For $1 < p \leq 2$, the numbers $d_{m,p}$ increase with m ; for $p \geq 2$, they are constant for $m \geq 1$. Therefore $d_{n-r,p} \leq M_{2^n,p}/2^n$. Averaging over the common coordinates and normalizing $\|F\|_p = 1$ proves the claim. $\square$

Theorem 1 (exact constants). *For Proposition 55 of the source paper, for every $n \geq 1$,*

$$c_n^p = \begin{cases} 2 - \frac{4 - 2^p}{2^n}, & 1 < p \leq 2, \\ 2^{p-1}, & 2 \leq p < \infty. \end{cases}$$

Consequently,

$$c = \sup_n c_n = \begin{cases} 2^{1/p}, & 1 < p \leq 2, \\ 2^{1-1/p}, & 2 \leq p < \infty. \end{cases}$$

Proof. The upper bounds follow from Lemmas 1 and 2, with $N = 2^n$.

For the lower bounds, choose disjoint A, B . If $1 < p \leq 2$, let F take the value $a = (2^{n-1})^{1/p}$ on one atom of $\{0, 1\}^n$, the value $-a$ on another atom, and 0 on all remaining atoms. Then $\|F\|_p = 1$, and for independent copies U, V of F ,

$$\mathbb{E}|U - V|^p = \frac{2(2a)^p + 4(2^n - 2)a^p}{2^{2n}} = 2 - \frac{4 - 2^p}{2^n}.$$

Thus c_n^p is at least the stated value.

If $p \geq 2$, let F be a Rademacher function depending on one coordinate. Then independent copies U, V satisfy $\mathbb{E}|U - V|^p = 2^{p-1}$. Hence $c_n^p \geq 2^{p-1}$, and the upper bound gives equality.

Taking the supremum over n gives the displayed value of c . $\square$

Verification Notes

The auxiliary script `code/check_finite_atom_conjecture.py` was used before the graph-spectrum identification to test the finite $1 < p < 2$ formula for small N . It is not part of the proof. The proof now rests on the conditioning reduction and on Amghibech’s complete-graph p -Laplacian spectrum as quoted in [3, Section 4.3].

Scope and Limitations

The exact constants c_n and c are settled. The remaining source question is whether the unit sphere $S_{L_p(\{0,1\}^\kappa)}$ is dichotomous. For $p = 2$, the earlier Hilbertian packet records the negative answer via the Erdős–Preiss and Preiss–Rodl theorem. For $p \neq 2$, the endpoint dichotomy problem is not resolved here.

Literature and Novelty Notes

The finite $1 < p < 2$ formula was first found here numerically, but the proof uses an existing graph p -Laplacian theorem of Amghibech. A web search for “complete graph p -Laplacian eigenvalues”, “largest eigenvalue graph p -Laplacian complete graph”, and exact pairwise-sum phrases led to Zhang’s arXiv:2110.06054 paper, whose Section 4.3 quotes the decisive Amghibech formula. I did not locate a freely downloadable copy of Amghibech’s original 2003 paper during the bounded search, so Zhang’s arXiv PDF is included as the supporting source for the quoted formula.

Human Review Recommendation

Reviewers should first check the factor $M_{N,p} = 2(N-1) \sup F_p$, since this is the only normalization-sensitive step. Then check that the integer maximization of Amghibech’s formula is correct in the two regimes $1 < p < 2$ and $p > 2$, and that the conditioning argument covers every admissible overlap pattern from Proposition 55.

References

- [1] P. Koszmider, *On Ramsey-type properties of the distance in nonseparable spheres*, arXiv:2308.07668.

- [2] S. Amghibech, *Eigenvalues of the discrete p -Laplacian for graphs*, Ars Combinatoria 67 (2003), 283–302.
- [3] D. Zhang, *Homological eigenvalues of graph p -Laplacians*, Journal of Topology and Analysis (2023), arXiv:2110.06054.
- [4] I. J. Schoenberg, *Metric spaces and positive definite functions*, Transactions of the American Mathematical Society 44 (1938), 522–536.